Arista Access Node WHI-ACC-61 QoS / Congestion Event Resolution SOP

Node: WHI-ACC-61 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_2020

Description

The failure domain of QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) in the Access role on the Arista vendor equipment at Whitefield East POP Node ID: WHI-ACC-61 can occur due to an intricate interplay of various network and hardware mechanics. This may involve bursty or uncontrolled traffic patterns, which can lead to unexpected Microbursts, interface tail drops, or QoS queue starvation. In the context of Arista's Extensible Operating System (EOS), the control plane manages the configuration and state of the data plane, which is responsible for forwarding packets. The data plane's forwarding decisions are influenced by the QoS configuration, which can be affected by various factors such as class-based policing, rate limiting, or traffic shaping. Furthermore, the presence of microbursts or sudden increases in traffic volume can lead to interface congestion, resulting in tail drops or queue starvation. Therefore, a comprehensive approach is necessary to troubleshoot and resolve these events, which involves examining the QoS configuration, data plane statistics, and other relevant metrics to identify the root cause of the issue.

Pre-requisite Checks

- Verify the Node ID: WHI-ACC-61 is accessible via the Arista EOS CLI.
- Check the redundant link status to ensure high availability.
- Dispatch remote hands to Whitefield East POP for physical access if required.
- Ensure the necessary permissions and access rights are granted for the Arista EOS CLI

Potential Root Causes

Cause	Probability
Misconfigured QoS policy leading to queue starvation	High
Microbursts or sudden increases in traffic volume causing interface congestion	Medium
Inadequate class-based policing or rate limiting	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos, Expected Output: Displays current QoS configuration and statistics, Description: This command helps identify any potential misconfigurations or issues with the QoS policy, which may be contributing to the congestion event.

Command: show interface detail, Expected Output: Displays detailed interface statistics and configuration, Description: This command provides insight into the interface's current state, including any
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congestion metrics, which may help isolate the root cause of the
issue.

Command: show process cpu, Expected Output: Displays CPU utilization
statistics, Description: This command helps identify any potential CPU
bottlenecks or spikes in CPU utilization, which may be related to the
congestion event.
```

Resolution Steps

Step 1: Step: Verify and Correct QoS Policy Configuration, Command: config qos policy (verify and correct policy configuration), Description: Review the QoS policy configuration to ensure it is correctly set up to handle the expected traffic patterns and correct any misconfigurations that may be contributing to the congestion event.

Step 2: Step: Adjust Class-Based Policing or Rate Limiting, Command: config qos policy class policing , Description: Adjust the class-based policing or rate limiting to ensure it is correctly set up to handle the expected traffic patterns and prevent congestion.

Step 3: Step: Verify and Adjust Interface Configuration, Command: config interface (verify and adjust interface configuration), Description: Review the interface configuration to ensure it is correctly set up to handle the expected traffic patterns and adjust any settings as necessary to prevent congestion.

Post-Check Verification

```
Command: show qos, Expected Output: Displays current QoS configuration
and statistics, Description: Verify that the QoS policy configuration
has been correctly updated and is functioning as expected.

Command: show interface detail, Expected Output: Displays detailed
interface statistics and configuration, Description: Verify that the
interface configuration has been correctly updated and is functioning
as expected.

Command: show process cpu, Expected Output: Displays CPU utilization
statistics, Description: Verify that CPU utilization is within
expected limits and that the congestion event has been resolved.
```

Roll-back Procedures

- Step: Restore Previous QoS Policy Configuration, Command: config qos policy rollback, Description: Restore the previous QoS policy configuration if the corrective actions fail to resolve the congestion event.

- Step: Restore Previous Interface Configuration, Command: config interface rollback, Description: Restore the previous interface configuration if the corrective actions fail to resolve the congestion event.